Multilingual multimodal recipe RAG: evaluation report

Baseline (original Nemotron VL tutorial) vs. improved pipeline vs. two experiments · Google Colab NVIDIA L4 (24GB) · 04 October 2026 · repository `github.com/ehzawad/multimodal-rag-nemotron`

1. Summary

0.400 → 0.748
Mean Recall@1 over 5 query sets, baseline full pipeline → final default

0.13 → 0.63
Mean Recall@1 on Spanish, Bengali, Chinese queries

14.9 s → 971 ms
Mean latency per query (retrieval + rerank)

We fixed the original tutorial notebook so it runs on a 24GB GPU, then built and evaluated three alternatives on a shared 1,000-query known-item benchmark (200 recipes × title, paraphrase, Spanish, Bengali and Chinese queries). The best pipeline, now the repository's `main`, replaces the Nemotron VL models with **Qwen3-VL-Embedding-2B** and **Qwen3-VL-Reranker-2B**, adds **BM25 fused with Reciprocal Rank Fusion only for queries whose words occur in the corpus vocabulary**, reranks on recipe text, and answers with **Qwen3.5-4B**. It is more accurate on every query set and an order of magnitude faster than the baseline's image + text reranking.

Pipeline	Stack	Title	Paraphrase	Spanish	Bengali	Chinese	Avg R@1	Latency
Baseline (original tutorial)	Nemotron dense → Nemotron rerank (image+text)	0.88	0.72	0.16	0.16	0.08	**0.400**	14.9 s
Improved	Nemotron dense + gated BM25 → Nemotron rerank (text)	0.98	0.86	0.48	0.32	0.28	**0.584**	589 ms
Exp: Qwen3-VL	Qwen3-VL dense + BM25 → Qwen3-VL rerank (text)	0.98	0.86	0.60	0.50	0.66	**0.720**	972 ms
Exp: Agentic	Qwen3.5-4B tool calls → Nemotron hybrid → rerank	0.94	0.78	0.54	0.44	0.48	**0.636**	5.3 s
Final (default)	Qwen3-VL dense + gated BM25 → Qwen3-VL rerank (text)	0.98	0.86	0.72	0.50	0.68	**0.748**	971 ms
Final + image rerank	… → Qwen3-VL rerank (image+text)	1.00	0.92	0.52	0.36	0.56	**0.672**	2.8 s
Final + agent	Qwen3.5-4B tool calls → final pipeline	0.94	0.80	0.60	0.48	0.52	**0.668**	5.6 s

Recall@1 of the target recipe. Rerank and agent rows use the first 50 queries per set (image + text rerank: 25), fast rows all 200; see §4 for n and confidence.

2. Task, data and hardware

- **Dataset:** `mrdbourke/recipe-synthetic-images-10k`, 10,096 Food.com recipes, each a rendered 1275×1650 recipe-card image plus markdown (title, description, ingredients, steps, tags, five reviews).
- **Task:** retrieve the right recipe for a text query (any language), an image, or both; optionally generate an answer grounded in the top three recipes.
- **Hardware:** one Google Colab NVIDIA L4 (24GB, 72W power cap), driven from the terminal with colab-persist over SSH. All final models together use 17.7GB of GPU memory, so the pipeline also fits an RTX A5000 (24GB).
- **One-off corpus embedding cost on the L4:** Qwen3-VL-Embedding-2B 48.5 min (image capped at ~400 tokens + recipe text without reviews); Nemotron Embed VL 2 h 19 min (tutorial format: 6 tiles + thumbnail and full text, ~4k tokens per recipe).

3. Pipelines compared

Branch	Retrieval	Rerank	Generation
Baseline	Nemotron Embed VL 1B (pinned commit 5b5ca69, transformers 4.57), dense only	Nemotron Rerank VL 1B, image + text, batched	Qwen3-VL-2B summary
Improved	Nemotron Embed VL (official revision 582e3bf, corpus re-embedded) + BM25, RRF; image queries described in text by the VLM	Nemotron Rerank VL (revision b8a9987), recipe text only	Qwen3.5-4B, grounded, cited
Exp: Qwen3-VL	Qwen3-VL-Embedding-2B (image + text documents) + BM25, RRF	Qwen3-VL-Reranker-2B	Qwen3.5-4B
Exp: Agentic	Qwen3.5-4B calls a `search_recipes` tool 1–3 times (translating / rewriting); each call runs the improved hybrid search; rankings fused with the original query's	Nemotron Rerank VL, text	Qwen3.5-4B, answers in the user's language
Final (main)	Qwen3-VL-Embedding-2B + BM25 **gated on query vocabulary coverage ≥ 0.8**, RRF	Qwen3-VL-Reranker-2B, text (image + text optional)	Qwen3.5-4B; agent optional

4. Evaluation method

Known-item retrieval on 200 recipes sampled with seed 42 (`results/eval_queries.json`). For each target recipe there are five queries: its **title**; a 5–12 word **paraphrase** written by Qwen3.5-4B from the recipe without the title's words; and the paraphrase machine-translated by Qwen3.5-4B into **Spanish**, **Bengali** and **Chinese**. We report Recall@1/@10 and MRR@10 of the target recipe and mean latency (retrieval, rerank and any LLM planning; answer generation excluded, ~20 s with Qwen3.5-4B).

Caveats: queries are synthetic; the Bengali translations are rough (many English food words transliterated, a few mistranslations); near-duplicate recipes exist (e.g. two "banana sour cream bread" recipes) so Recall@1 under-counts; slow methods were evaluated on 50 (rerank, agent) or 25 (image + text rerank, thinking agent) queries per set, giving roughly ±7–10 points of sampling error, so differences smaller than that between slow methods are not significant.

5. Results

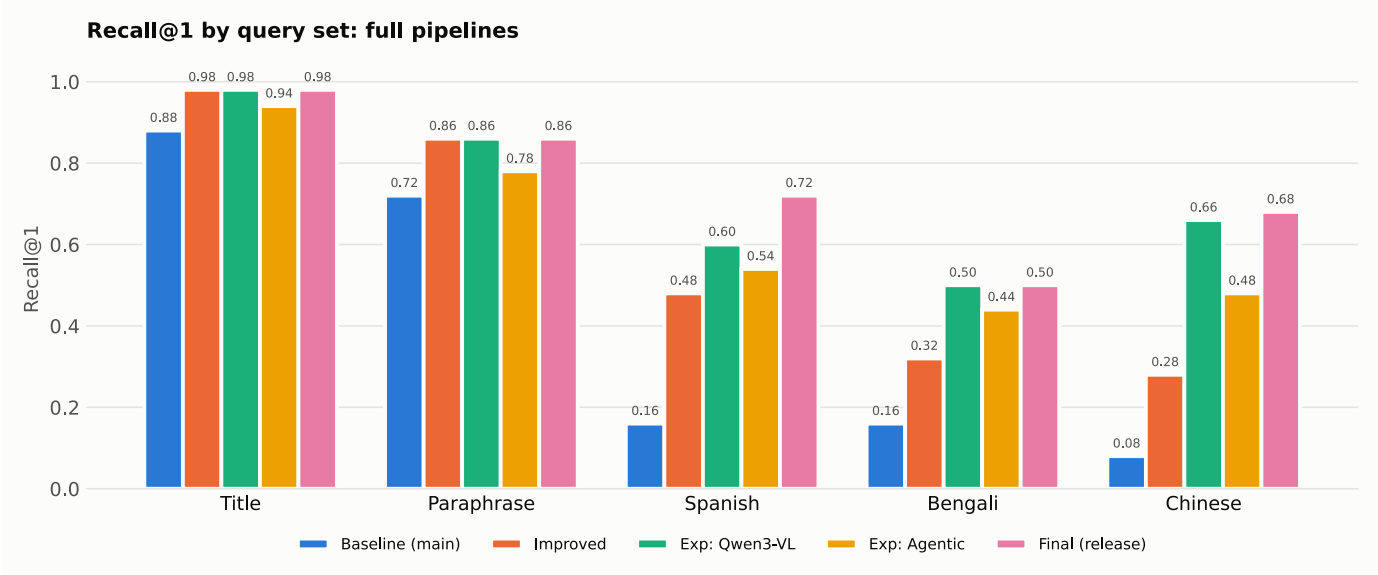


Figure 1. Recall@1 per query set for the representative full pipeline of each branch.

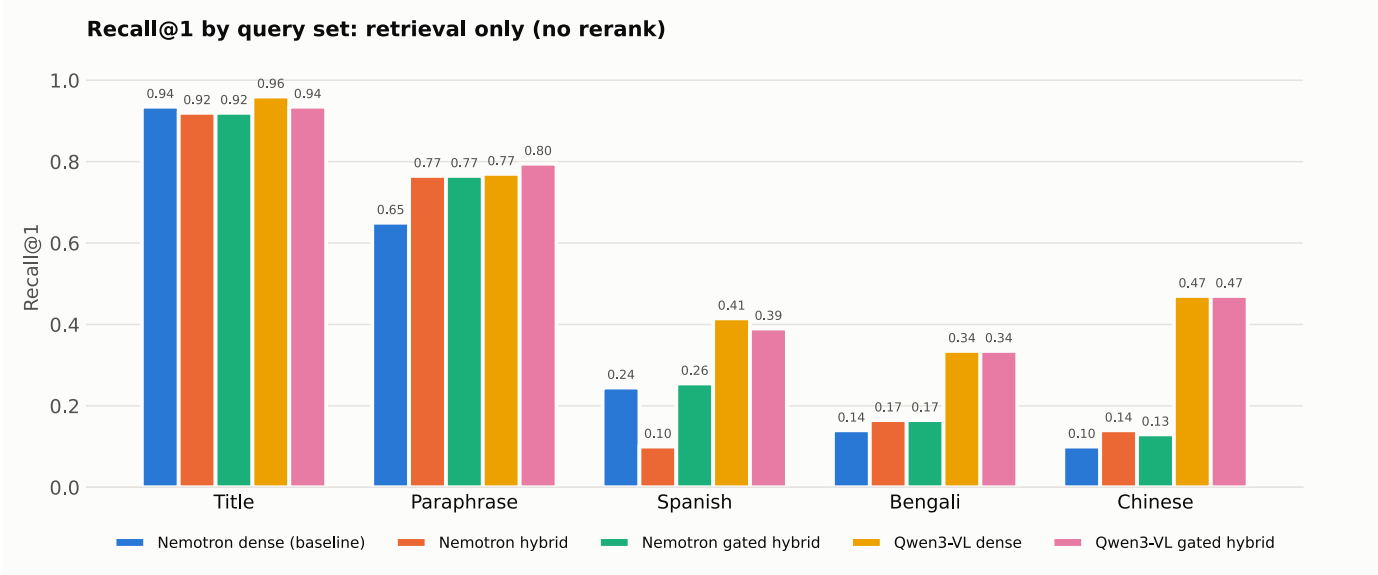


Figure 2. Retrieval only (no rerank): the embedding model is the main lever for non-English queries; ungated BM25 hurts Spanish.

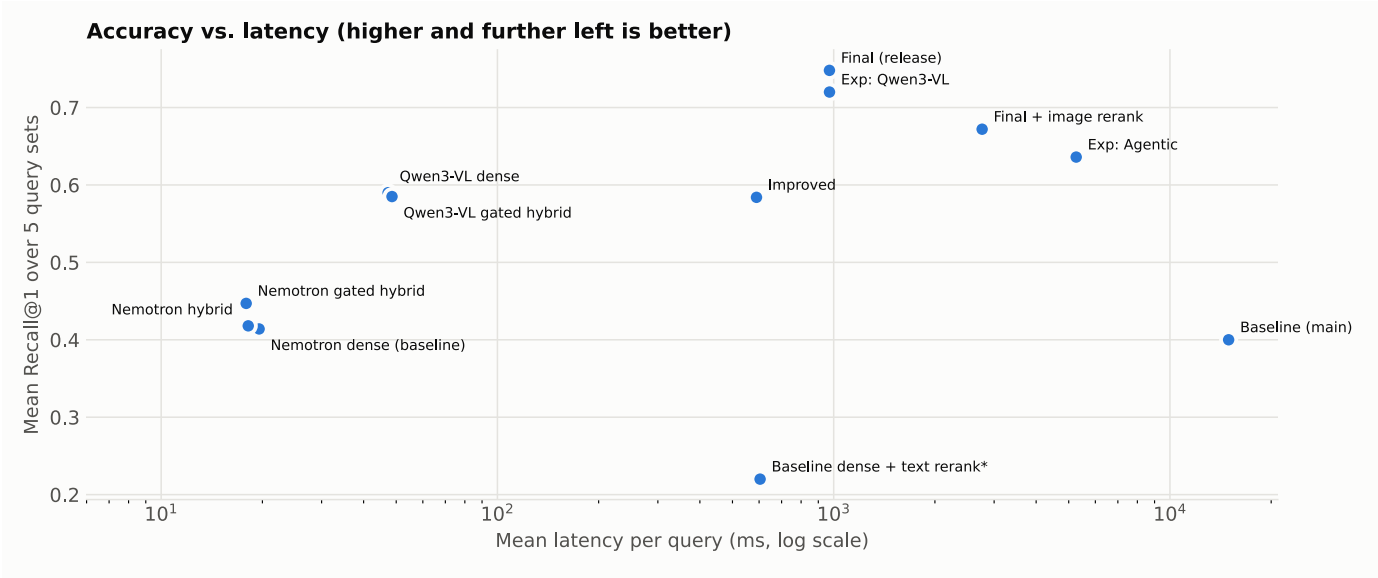


Figure 3. Mean Recall@1 against mean latency per query. *Baseline stack with text-only rerank exposes the old reranker's text-only failure.

Recall@1 (all runs and methods)

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Baseline stack (pinned commits, transformers 4.57)						
baseline: dense	0.935	0.650	0.245	0.140	0.100	**0.414**
baseline: dense + rerank (image+text)	0.880	0.720	0.160	0.160	0.080	**0.400**
bm25 only	0.820	0.690	0.015	0.005	0.005	**0.307**
improved: hybrid (dense + bm25, RRF)	0.895	0.735	0.110	0.140	0.105	**0.397**
improved: hybrid (bm25 weight 0.5)	0.905	0.735	0.115	0.140	0.105	**0.400**
improved: hybrid + rerank (text)	0.480	0.320	0.100	0.120	0.080	**0.220**
Improved stack (official Nemotron revisions, re-embedded)						
baseline: dense	0.965	0.750	0.260	0.165	0.130	**0.454**
bm25 only	0.820	0.690	0.015	0.005	0.005	**0.307**
improved: hybrid (dense + bm25, RRF)	0.920	0.765	0.100	0.165	0.140	**0.418**
improved: gated hybrid (bm25 if coverage >= 0.8)	0.920	0.765	0.255	0.165	0.130	**0.447**

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
improved: hybrid + rerank (text)	0.980	0.860	0.460	0.320	0.280	**0.580**
improved: gated hybrid + rerank (text)	0.980	0.860	0.480	0.320	0.280	**0.584**
Experiment: Qwen3-VL embedding + reranker						
qwen3-vl: dense	0.960	0.770	0.415	0.335	0.470	**0.590**
qwen3-vl: hybrid (dense + bm25, RRF)	0.935	0.795	0.150	0.335	0.465	**0.536**
qwen3-vl: hybrid + qwen3-vl rerank (text)	0.980	0.860	0.600	0.500	0.660	**0.720**
qwen3-vl: hybrid + qwen3-vl rerank (image+text)	1.000	0.920	0.560	0.360	0.480	**0.664**
Experiment: agentic (Qwen3.5-4B tool calls) on Nemotron						
agentic: tool-call rewrite + hybrid	0.780	0.540	0.280	0.400	0.320	**0.464**
agentic: tool-call rewrite + hybrid + rerank (text)	0.940	0.780	0.540	0.440	0.480	**0.636**
agentic (thinking): rewrite + hybrid + rerank (text)	0.920	0.760	0.520	0.440	0.480	**0.624**
Final release: Qwen3-VL + gated hybrid						
final: qwen3-vl gated hybrid	0.935	0.795	0.390	0.335	0.470	**0.585**
final: qwen3-vl gated hybrid + qwen3-vl rerank (text)	0.980	0.860	0.720	0.500	0.680	**0.748**
final: qwen3-vl gated hybrid + qwen3-vl rerank (image+text)	1.000	0.920	0.520	0.360	0.560	**0.672**
final: agent + qwen3-vl gated hybrid + qwen3-vl rerank (text)	0.940	0.800	0.600	0.480	0.520	**0.668**

MRR@10

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Baseline stack (pinned commits, transformers 4.57)						
baseline: dense	0.963	0.759	0.344	0.202	0.176	**0.489**
baseline: dense + rerank (image+text)	0.940	0.797	0.265	0.185	0.176	**0.472**
bm25 only	0.885	0.772	0.029	0.005	0.008	**0.340**
improved: hybrid (dense + bm25, RRF)	0.937	0.820	0.177	0.202	0.181	**0.463**
improved: hybrid (bm25 weight 0.5)	0.945	0.820	0.229	0.202	0.180	**0.475**
improved: hybrid + rerank (text)	0.618	0.421	0.176	0.151	0.115	**0.296**
Improved stack (official Nemotron revisions, re-embedded)						
baseline: dense	0.983	0.830	0.379	0.221	0.207	**0.524**
bm25 only	0.885	0.772	0.029	0.005	0.008	**0.340**
improved: hybrid (dense + bm25, RRF)	0.952	0.839	0.180	0.221	0.212	**0.481**
improved: gated hybrid (bm25 if coverage >= 0.8)	0.952	0.839	0.362	0.221	0.207	**0.516**
improved: hybrid + rerank (text)	0.990	0.903	0.492	0.335	0.308	**0.606**
improved: gated hybrid + rerank (text)	0.990	0.903	0.518	0.335	0.308	**0.611**
Experiment: Qwen3-VL embedding + reranker						
qwen3-vl: dense	0.979	0.851	0.533	0.440	0.587	**0.678**
qwen3-vl: hybrid (dense + bm25, RRF)	0.959	0.864	0.269	0.440	0.583	**0.623**
qwen3-vl: hybrid + qwen3-vl rerank (text)	0.990	0.903	0.649	0.572	0.750	**0.773**
qwen3-vl: hybrid + qwen3-vl rerank (image+text)	1.000	0.940	0.611	0.487	0.613	**0.730**
Experiment: agentic (Qwen3.5-4B tool calls) on Nemotron						
agentic: tool-call rewrite + hybrid	0.864	0.646	0.416	0.461	0.440	**0.565**
agentic: tool-call rewrite + hybrid + rerank (text)	0.967	0.851	0.630	0.521	0.611	**0.716**
agentic (thinking): rewrite + hybrid + rerank (text)	0.938	0.839	0.600	0.545	0.584	**0.701**

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Final release: Qwen3-VL + gated hybrid						
final: qwen3-vl gated hybrid	0.959	0.864	0.501	0.440	0.587	**0.670**
final: qwen3-vl gated hybrid + qwen3-vl rerank (text)	0.990	0.903	0.759	0.572	0.770	**0.799**
final: qwen3-vl gated hybrid + qwen3-vl rerank (image+text)	1.000	0.940	0.620	0.487	0.673	**0.744**
final: agent + qwen3-vl gated hybrid + qwen3-vl rerank (text)	0.965	0.872	0.687	0.572	0.647	**0.749**

Recall@10

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Baseline stack (pinned commits, transformers 4.57)						
baseline: dense	1.000	0.945	0.590	0.355	0.370	**0.652**
baseline: dense + rerank (image+text)	1.000	0.960	0.560	0.240	0.360	**0.624**
bm25 only	0.980	0.940	0.075	0.005	0.015	**0.403**
improved: hybrid (dense + bm25, RRF)	0.995	0.955	0.410	0.355	0.380	**0.619**
improved: hybrid (bm25 weight 0.5)	0.995	0.960	0.520	0.355	0.380	**0.642**
improved: hybrid + rerank (text)	0.940	0.700	0.380	0.300	0.220	**0.508**
Improved stack (official Nemotron revisions, re-embedded)						
baseline: dense	1.000	0.960	0.600	0.360	0.380	**0.660**
bm25 only	0.980	0.940	0.075	0.005	0.015	**0.403**
improved: hybrid (dense + bm25, RRF)	0.995	0.965	0.485	0.360	0.380	**0.637**
improved: gated hybrid (bm25 if coverage >= 0.8)	0.995	0.965	0.565	0.360	0.380	**0.653**
improved: hybrid + rerank (text)	1.000	0.960	0.540	0.360	0.360	**0.644**
improved: gated hybrid + rerank (text)	1.000	0.960	0.580	0.360	0.360	**0.652**
Experiment: Qwen3-VL embedding + reranker						
qwen3-vl: dense	1.000	0.980	0.775	0.670	0.835	**0.852**
qwen3-vl: hybrid (dense + bm25, RRF)	0.995	0.980	0.625	0.670	0.840	**0.822**
qwen3-vl: hybrid + qwen3-vl rerank (text)	1.000	0.960	0.740	0.700	0.900	**0.860**
qwen3-vl: hybrid + qwen3-vl rerank (image+text)	1.000	0.960	0.720	0.760	0.880	**0.864**
Experiment: agentic (Qwen3.5-4B tool calls) on Nemotron						
agentic: tool-call rewrite + hybrid	1.000	0.880	0.720	0.660	0.740	**0.800**
agentic: tool-call rewrite + hybrid + rerank (text)	1.000	0.940	0.780	0.700	0.820	**0.848**
agentic (thinking): rewrite + hybrid + rerank (text)	1.000	0.960	0.720	0.760	0.760	**0.840**
Final release: Qwen3-VL + gated hybrid						
final: qwen3-vl gated hybrid	0.995	0.980	0.735	0.670	0.835	**0.843**
final: qwen3-vl gated hybrid + qwen3-vl rerank (text)	1.000	0.960	0.840	0.700	0.920	**0.884**
final: qwen3-vl gated hybrid + qwen3-vl rerank (image+text)	1.000	0.960	0.840	0.760	0.920	**0.896**
final: agent + qwen3-vl gated hybrid + qwen3-vl rerank (text)	1.000	0.980	0.880	0.780	0.860	**0.900**

Mean latency per query (ms)

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Baseline stack (pinned commits, transformers 4.57)						
baseline: dense	19	19	19	20	20	**20**
baseline: dense + rerank (image+text)	16892	16019	15640	12213	13963	**14945**

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
bm25 only	1	0	1	0	0	**0**
improved: hybrid (dense + bm25, RRF)	20	20	21	20	20	**20**
improved: hybrid (bm25 weight 0.5)	20	20	20	20	20	**20**
improved: hybrid + rerank (text)	562	570	611	689	589	**604**
Improved stack (official Nemotron revisions, re-embedded)						
baseline: dense	17	18	18	18	17	**18**
bm25 only	1	0	1	0	0	**0**
improved: hybrid (dense + bm25, RRF)	18	18	19	18	18	**18**
improved: gated hybrid (bm25 if coverage >= 0.8)	18	18	18	17	18	**18**
improved: hybrid + rerank (text)	553	546	603	683	579	**593**
improved: gated hybrid + rerank (text)	544	552	587	683	581	**589**
Experiment: Qwen3-VL embedding + reranker						
qwen3-vl: dense	47	48	47	47	47	**47**
qwen3-vl: hybrid (dense + bm25, RRF)	48	48	48	48	48	**48**
qwen3-vl: hybrid + qwen3-vl rerank (text)	935	935	969	1060	959	**972**
qwen3-vl: hybrid + qwen3-vl rerank (image+text)	2708	2722	2709	2825	2703	**2733**
Experiment: agentic (Qwen3.5-4B tool calls) on Nemotron						
agentic: tool-call rewrite + hybrid	4365	4893	4464	4538	5273	**4707**
agentic: tool-call rewrite + hybrid + rerank (text)	4906	5409	5027	5117	5854	**5263**
agentic (thinking): rewrite + hybrid + rerank (text)	14152	14773	16136	17365	19527	**16391**
Final release: Qwen3-VL + gated hybrid						
final: qwen3-vl gated hybrid	48	49	48	49	49	**49**
final: qwen3-vl gated hybrid + qwen3-vl rerank (text)	932	935	975	1057	958	**971**
final: qwen3-vl gated hybrid + qwen3-vl rerank (image+text)	2708	2732	2766	2873	2738	**2763**
final: agent + qwen3-vl gated hybrid + qwen3-vl rerank (text)	5307	5796	5410	5477	6197	**5637**

Queries evaluated (n)

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Baseline stack (pinned commits, transformers 4.57)						
baseline: dense	200	200	200	200	200	**200**
baseline: dense + rerank (image+text)	25	25	25	25	25	**25**
bm25 only	200	200	200	200	200	**200**
improved: hybrid (dense + bm25, RRF)	200	200	200	200	200	**200**
improved: hybrid (bm25 weight 0.5)	200	200	200	200	200	**200**
improved: hybrid + rerank (text)	50	50	50	50	50	**50**
Improved stack (official Nemotron revisions, re-embedded)						
baseline: dense	200	200	200	200	200	**200**
bm25 only	200	200	200	200	200	**200**
improved: hybrid (dense + bm25, RRF)	200	200	200	200	200	**200**
improved: gated hybrid (bm25 if coverage >= 0.8)	200	200	200	200	200	**200**
improved: hybrid + rerank (text)	50	50	50	50	50	**50**
improved: gated hybrid + rerank (text)	50	50	50	50	50	**50**

Method	Title	Paraphrase	Spanish	Bengali	Chinese	Avg
Experiment: Qwen3-VL embedding + reranker						
qwen3-vl: dense	200	200	200	200	200	**200**
qwen3-vl: hybrid (dense + bm25, RRF)	200	200	200	200	200	**200**
qwen3-vl: hybrid + qwen3-vl rerank (text)	50	50	50	50	50	**50**
qwen3-vl: hybrid + qwen3-vl rerank (image+text)	25	25	25	25	25	**25**
Experiment: agentic (Qwen3.5-4B tool calls) on Nemotron						
agentic: tool-call rewrite + hybrid	50	50	50	50	50	**50**
agentic: tool-call rewrite + hybrid + rerank (text)	50	50	50	50	50	**50**
agentic (thinking): rewrite + hybrid + rerank (text)	25	25	25	25	25	**25**
Final release: Qwen3-VL + gated hybrid						
final: qwen3-vl gated hybrid	200	200	200	200	200	**200**
final: qwen3-vl gated hybrid + qwen3-vl rerank (text)	50	50	50	50	50	**50**
final: qwen3-vl gated hybrid + qwen3-vl rerank (image+text)	25	25	25	25	25	**25**
final: agent + qwen3-vl gated hybrid + qwen3-vl rerank (text)	50	50	50	50	50	**50**

6. Findings

6.1 The original notebook did not run, and had latent bugs

- `ValueError: Sample larger than population`: image paths were globbed from a local folder that only existed on the author's machine (0 images). Fixed by sampling the dataset's `image` column.
- Reranking 20 image + text pairs in one forward pass runs out of memory on a 24GB GPU (CUDA OOM at 20.4GB) → batched (4 pairs).
- The reranker's processor kwargs were built under a misspelled name and never passed; image queries were passed to the reranker as its text `question`; `get_top_n_scores_and_samples` ignored `top_n`; "Run all" blocked on `demo.launch(debug=True)` and would have tried to publish a Space. All fixed (`notebooks/multimodal_rag_tutorial_baseline.ipynb`, end to end on the L4 in 138 s).

6.2 Hidden embedding-compatibility trap

The demo Space's precomputed document embeddings only reproduce with the tutorial's pinned embed commit (cosine 0.9999). Every official revision of `llama-nemotron-embed-vl-1b-v2` tokenizes and masks differently (cosine ≈ 0.86, under both transformers 4.57 and 5.x), and the pinned commit cannot run on transformers 5, which Qwen3.5 requires. Re-embedding the corpus with the official revision fixed the mismatch *and* improved plain dense retrieval: title Recall@1 0.94 → 0.96, paraphrase 0.65 → 0.75.

6.3 The reranker revision matters more than image inputs

With the tutorial's pinned reranker commit, image + text reranking is *worse* than no reranking on most sets (mean Recall@1 0.400 vs 0.414) at ~15 s per query, and text-only reranking is broken (title Recall@1 0.48). The current official revision reranks recipe text well in ~0.6 s (title 0.98, paraphrase 0.86). Dropping the reviews from the reranked text halves its cost.

6.4 Hybrid search helps English and hurts other languages: gate it

BM25 over the English corpus improves English paraphrase retrieval but injects noise for other languages; with Qwen3-VL, ungated hybrid drops Spanish Recall@1 from 0.41 to 0.15. A free, language-agnostic gate fixes this: use BM25 only if ≥ 80% of the query's non-stopword tokens occur in the corpus vocabulary. On our queries this passes

100% of English, 9% of Spanish and ~0% of Bengali/Chinese queries, keeping the English gains and restoring dense-only behaviour elsewhere (Nemotron Spanish: 0.10 ungated vs 0.26 gated).

6.5 A multilingual multimodal embedding model is the biggest single lever

Qwen3-VL-Embedding-2B improves dense Recall@1 on non-English queries 1.6–3.6× versus the re-embedded Nemotron (Spanish 0.26 → 0.41, Bengali 0.17 → 0.34, Chinese 0.13 → 0.47) while matching it on English, and its reranker (with the BM25 gate) adds another 16–30 points on every non-English set.

6.6 Agentic search planning helps weak retrievers but does not win

Letting Qwen3.5-4B translate/rewrite the request via tool calls lifts the Nemotron pipeline on Bengali (dense 0.17 → agent 0.40) but hurts exact-title queries (rewriting loses the exact name), costs ~5 s per query with the reference kernels, and thinking mode adds ~10 s more with no accuracy gain. On top of the Qwen3-VL pipeline it does not improve the average (0.668 with agent vs 0.748 without), so it ships as an option (`--agent`) for vague or multi-part requests rather than the default.

6.7 Image + text reranking: best for English only

Reranking with the recipe-card images reaches the best English numbers (title 1.00, paraphrase 0.92) but is worse than text-only reranking for non-English queries and ~3× slower; text reranking is the default and image + text an option (`--rerank-images`).

6.8 Generation

Qwen3.5-4B (natively multimodal, Feb 2026) produces grounded, cited markdown answers and describes image queries for lexical search (e.g. the kitchen-bench photo → "tomato pasta with fresh basil and parmesan cheese"). Thinking is disabled for latency; an answer takes ~20 s on the L4 because `causal-conv1d` is not installed (flash-linear-attention is).

7. Recommendation (what is on main)

- Default: Qwen3-VL-Embedding-2B dense + vocabulary-gated BM25 (RRF) → Qwen3-VL-Reranker-2B on recipe text → Qwen3.5-4B grounded answer.
- Options: `--rerank-images` for maximum English accuracy, `--agent` for vague requests, `--backend nemotron` for the improved Nemotron stack. The fixed original tutorial is kept as `notebooks/multimodal_rag_tutorial_baseline.ipynb` / `standalone/baseline_app.py`.
- Future work: human-labelled multi-answer relevance judgements; native-speaker queries (especially Bengali); install causal-conv1d or serve Qwen3.5 with vLLM for faster answers; Qwen3-VL-Embedding-8B on a 48GB GPU; host the 41MB corpus embeddings on the Hub.

8. Reproduce

`pip install -r requirements.txt` · `python scripts/build_qwen_vl_embeddings.py` (corpus, ~50 min) · `python scripts/eval_final.py` · baselines: `MMRAG_NEMOTRON_REVISIONS=original` with transformers 4.57.3 (`requirements-baseline.txt`) and `python scripts/eval_retrieval.py` · `python scripts/eval_qwen_vl.py` · `python scripts/eval_agentic.py` · `python scripts/make_report_charts.py && python scripts/make_report.py`.